

Munki Jeong

Email: jmunki1004@gmail.com Citizenship: Republic of Korea

Research Interests

Developing mathematical and computational frameworks relevant to:

- Dynamic game theory (in both evolutionary and mean-field game frameworks)
- Stochastic optimal control
- Complex systems: emergent and nonlinear behavior in large-scale interacting dynamical systems

Education

Gwangju Institute of Science and Technology (GIST) Sp 2019 – Sp 2026 (expected)
(Including 2 years of leave for mandatory military service)
B.S. in Electrical Engineering and Computer Science
B.S. in Mathematical Science
GPA: 3.95 / 4.5
Relevant coursework: Linear Algebra, Ordinary Differential Equations, Probability and Statistics, Abstract Algebra, Graph Theory, Data Structures, Introduction to Algorithms, Machine Learning and Deep Learning, Signals and Systems

University of California, Berkeley Spring 2024 – Spring 2025
Visiting Undergraduate on Scholarships (Berkeley Global Access Program and Summer Session)
GPA: 3.838 / 4.0
Relevant coursework: Math 202A (General Topology and Measure Theory), Math 105 (Analysis II), Math 221 (Numerical Linear Algebra), Math 128B (Numerical Analysis II), EE 227 (Convex Optimization), EE 223 (Stochastic Systems and Control), Stat 134 (Mathematical Statistics), Stat 155 (Game Theory), Econ 104 (Econometrics)

Korea Advanced Institute of Science and Technology (KAIST) Fall 2025
Undergraduate Exchange
Relevant coursework: MAS 550 (Measure-Theoretic Probability Theory)

Research Experience

Individual Research, UC Berkeley, Department of Statistics Aug 2024 – Present
Advisor: Prof. Alexander Strang
Title (working paper): *The Emergence of Hierarchy in the Steady-States of Populations Evolving in Response to Random Payouts: A Local Analysis*

Undergraduate Thesis, GIST, Department of Mathematical Science
Advisor: Prof. Geonho Hwang

Sep 2025 – Present

- Study of functional-analytic and computational aspects of approximation methods for explicit solutions of multiparametric linear programming
- Initially began as a convex optimization course project (EE 227BT, Fall 2024), where the high-level idea was first suggested by Prof. Benjamin Recht (UC Berkeley, EECS)

Course Project, UC Berkeley, Department of Mathematics

Oct 2024 – Dec 2024

Course: Math 221 (Graduate Numerical Linear Algebra)

Instructor: Prof. James Demmel

Collaborator: Ainsley Clark (Ph.D. student, UC Berkeley Mathematics)

Title: *The Interacting Particles Problem as a Structured Matrix-Vector Multiplication*

- Proposed and implemented a fast method to compute the Gauss transform based on the Fast Fourier Transform
- Empirically compared the proposed method with the Fast Gauss Transform

Publications & Manuscripts

- Jeong, M. & Strang, A. (preprint, which must be available at the time of application). *The Emergence of Hierarchy in the Steady-States of Populations Evolving in Response to Random Payouts: A Local Analysis*.

Talks & Presentations

Slide Presentation at Solid Reading Group

Apr 2025

“Payoff Dynamics Model and Evolutionary Dynamics Model: Feedback and Convergence to Equilibria (Park, C. Martins, and S. Shamma)”

Poster Presentation for Math 221 Final Project

Dec 2024

“The Interacting Particles Problem as a Structured Matrix-Vector Multiplication”

Seminars & Reading Groups

Solid Reading Group (Game Theory), UC Berkeley EECS

Feb 2025 – May 2025

- Engaged in discussions with graduate students, postdocs, and faculty members (mostly with engineering backgrounds) on foundations and applications of *Population Game Theory*

Vecchia Approximation & Fast Multipole Method Reading Group, UC Berkeley

Jul 2025 – Oct 2025

- Reading group led by Prof. Alexander Strang

Teaching & Mentoring

- Teaching Assistant, GIST: Linear Algebra and Applications (Fall 2023)

Awards & Honors

- BGA Visiting Student Scholarship, UC Berkeley Extension (Fall 2024)
- Study Abroad Program Scholarship, GIST (Spring 2024)
- Korea Government Scholarship (8 semesters in total)
- Academic Excellence Scholarship, GIST (Spring 2020, Spring 2023)

Programming Skills

Python, MATLAB, R, $\text{\LaTeX}$

Languages

- Korean: Native
- English: Fluent (TOEFL iBT: 102/120)